

MUONIO SAMMALTUNTURI, Finland

WMO: 028210

Lat: 67.973N

Lon: 24.116E

Elev: 566

StdP: 94.71

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.7	-20.3	-25.1	0.4	-22.6	-22.4	0.5	-19.6	19.6	-9.4	16.8	-7.4	7.8	80	0.972

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.7	22.3	14.6	20.0	13.7	18.0	12.6	16.0	20.4	14.6	18.5	13.5	16.8	4.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.5	11.0	17.1	13.0	10.0	15.3	12.0	9.3	14.4	46.6	20.6	42.7	18.7	39.5	16.9	20.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.6	13.9	12.5	-24.2	23.5	3.1	2.1	-26.4	25.0	-28.3	26.3	-30.0	27.5	-32.3	29.0		
			-24.4	17.1	2.9	1.9	-26.5	18.5	-28.3	19.7	-29.9	20.7	-32.0	22.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.3	-11.0	-10.4	-7.3	-2.1	3.9	8.5	12.6	10.4	5.9	-1.3	-5.6	-8.2
	DBStd	9.03	4.63	5.29	3.73	3.62	4.91	4.39	4.16	3.62	3.22	3.59	3.57	4.14
	HDD10.0	3970	650	572	536	364	202	80	19	38	126	351	469	563
	HDD18.3	6819	908	805	794	614	449	296	184	246	373	609	719	822
	CDD10.0	201	0	0	0	0	12	34	101	51	3	0	0	0
	CDD18.3	9	0	0	0	0	1	1	7	1	0	0	0	0
	CDH23.3	24	0	0	0	0	1	2	19	3	0	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	6.9	7.8	7.8	7.8	6.8	6.6	6.4	6.0	5.8	6.3	6.6	7.2	8.0
Precipitation	PrecAvg	592	40	34	29	33	39	63	85	71	55	49	52	43
	PrecMax	758	78	76	61	75	96	106	141	144	112	93	96	76
	PrecMin	443	9	13	9	10	8	9	32	33	14	13	14	9
	PrecStd	92	15	16	14	16	20	23	30	28	27	21	21	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.4	2.0	2.0	10.9	21.7	23.0	25.6	22.7	14.5	7.3	4.0	0.9
		MCWB	-0.7	1.3	-0.4	5.3	10.8	14.7	15.9	16.1	10.4	6.0	3.4	0.4
	2%	DB	-1.9	-1.3	0.3	6.8	17.3	19.7	23.1	20.0	12.9	5.9	1.7	-0.7
		MCWB	-2.4	-1.7	-1.2	3.1	10.1	13.1	14.9	14.0	10.3	5.2	1.1	-1.0
	5%	DB	-3.3	-2.4	-1.0	4.5	14.2	17.4	21.1	17.6	11.6	4.9	0.0	-1.8
		MCWB	-3.6	-2.6	-2.0	1.7	8.5	11.6	14.2	13.1	9.4	4.3	-0.2	-2.0
	10%	DB	-4.6	-3.6	-2.2	2.8	10.8	15.2	19.0	15.5	10.2	3.5	-1.0	-2.7
		MCWB	-4.9	-3.9	-2.9	0.3	6.4	10.1	13.7	11.9	8.6	3.0	-1.2	-2.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.7	1.4	0.2	5.8	12.2	16.8	17.7	17.9	12.4	6.6	3.6	0.5
		MCDB	-0.5	2.3	1.2	10.4	18.5	21.9	20.9	22.0	13.0	6.8	3.9	0.8
	2%	WB	-2.3	-1.6	-0.9	3.6	10.5	14.3	16.3	15.1	11.0	5.4	1.1	-0.9
		MCDB	-2.0	-1.3	0.0	6.3	16.5	18.2	20.7	18.0	12.0	5.8	1.5	-0.7
	5%	WB	-3.6	-2.7	-1.9	2.0	8.9	12.6	15.2	13.6	9.9	4.3	-0.2	-2.0
		MCDB	-3.3	-2.5	-1.2	4.3	13.8	15.4	19.5	16.3	11.1	4.8	0.1	-1.8
	10%	WB	-4.9	-3.9	-2.9	0.7	6.8	10.9	14.2	12.5	9.0	3.0	-1.2	-2.9
		MCDB	-4.6	-3.6	-2.3	2.4	10.3	14.7	18.4	14.8	9.9	3.5	-1.1	-2.7
Mean Daily Temperature Range	5% DB	MDBR	4.5	4.2	4.5	5.5	6.5	6.8	6.7	5.9	4.3	3.3	3.8	3.9
		MCDBR	5.7	4.5	5.2	7.5	9.7	9.2	8.4	7.8	5.6	4.0	4.2	4.0
		MCWBR	5.8	4.4	4.3	4.5	5.2	4.9	4.0	3.9	3.7	3.3	4.2	3.9
	5% WB	MCDBR	5.7	4.4	4.6	7.1	9.0	8.1	7.4	6.9	4.5	3.7	4.3	3.9
		MCWBR	5.9	4.3	4.1	4.5	5.2	4.9	4.2	3.9	3.5	3.3	4.3	3.9
Clear-Sky Solar Irradiance	taub	0.176	0.234	0.270	0.299	0.317	0.321	0.330	0.315	0.287	0.254	N/A	N/A	
	taud	1.917	2.106	2.213	2.244	2.447	2.515	2.500	2.521	2.572	2.338	N/A	N/A	
	Ebn at Noon	125	578	774	851	880	885	863	839	782	569	N/A	0	
	Edh at Noon	9	46	76	99	92	88	87	76	56	37	N/A	0	
All-Sky Solar Radiation	RadAvg	0.04	0.50	2.00	3.94	4.89	4.98	4.66	3.37	1.78	0.59	0.10	0.00	
	RadStd	0.00	0.06	0.12	0.20	0.48	0.36	0.46	0.26	0.20	0.10	0.01	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page